

Mainak Majumder, M. Sc.

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📍 Linz, Austria

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Work Experiences

- 02/2022 – present  **University Assistant**, LIT Cyber-Physical Systems Lab
Johannes Kepler Universität Linz, Austria
- 04/2021 – 01/2022  **Project Assistant**, LIT Cyber-Physical Systems Lab
Johannes Kepler Universität Linz, Austria
- 10/2018 – 02/2021  **Research Assistant**, Institute for Industrial IT (inIT)
Technische Hochschule Ostwestfalen-Lippe, Lemgo, Germany
- 03/2018 – 08/2018  **Student Intern**, ABB AG Forschungszentrum, Ladenburg, Germany

Education

- 03/2022 – 03/2026  **PhD, Computer Science**
Johannes Kepler Universität Linz, Austria
Topics: *AAS, CPPS, Information Modelling, OPC UA, RAMi 4.0*
- 10/2014 – 07/2018  **M. Sc, Electrical Engineering and Information Technology**
Otto von Guericke Universität (OVGU), Magdeburg, Germany
Topics: *Information Modelling, OPC UA, RDF, Semantic Web* (Grade: 2.1)
- 08/2009 – 08/2013  **B. Tech, Electrical Engineering**
West Bengal University of Technology (WBUT), Kolkata, India (Grade: 1.8)

Technical Competencies

Information Technology (IT)

- Programming  C, Java
- Modelling  UML, EMF, RDF
- Paradigm  MDA, DDD
- IDE  Eclipse, IntelliJ IDEA, Visual Studio Code
- Network  TCP/IP, HTTP(S), MQTT, Kafka, Wireshark
- Tools  Node-RED, Git, CMake

Operational Technology (OT)

- Programming  Structured Text, Function Block Diagram (IEC 61131-3), IEC 61499 FB
- Modelling  UaModeler (OPC UA), AutomationML, Asset Administration Shell
- IDE  Eclipse 4diac, CODESYS
- Network  OPC UA, Modbus/TCP

Teaching Experiences

Lecture	 Production Automation Systems Lecture , JKU Topic: <i>OPC UA</i>
Practicum	 Network Embedded Systems , JKU Topics: <i>ELITeBoard, STM32, GPIO, Timer, I2C, SPI, UART</i>
	 Projects in Network Embedded Systems , JKU Topics: <i>Arduino, Raspberry PI, MQTT</i>
	 Production Automation Systems , JKU Topics: <i>IEC 61131-3, CODESYS, IEC 61499, Eclipse 4diac, FactoryIO</i>
	 Systems Programming , JKU Topics: <i>C</i>
	 Computer Networks , TH OWL Topics: <i>OSI Model, TCIP/IP, Ethernet</i>
Bachelor Theses	 Evaluation of a PLC4X-Based Data Collection Architecture
	 Visualization of OPC UA Information Model
	 Integrating OPC UA with Apache Kafka
	 Implementing Control Application Based on Model-Driven Design
	 Optimizing Solar Power Consumption Using AI Algorithms

Projects

FIND	 Future Industrial Network Architecture  Topic: <i>Software-defined Network, OpenFlow, SNMP, LLDP, YANG</i>
iAP	 Industrial Automation Platform for Big Data  Topic: <i>OPC UA, MQTT, Docker, Modbus</i>
BrAIN	 Brownfield Artificial Intelligence Network for Forging of High-Quality Aerospace Components  Topic: <i>OPC UA, Asset Administration Shell</i>

Miscellaneous Activities

Attendee	 EIT Digital Summer School Programme 2022 , TU Munich Topic: <i>IoT Platforms for Industry 4.0</i>
Speaker	 1st 4 days of Eclipse 4diac Winter School Topic: <i>Unlocking the IT/OT Convergence with OPC UA</i>
Reviewer	 ETFA 22, 23, 24 <i>Track 1: Information Technology in Automation</i> <i>WIP01: Information Technology in Automation</i> <i>SS04: Capability- and Skill-based Engineering of Manufacturing Systems</i>
	 IECON 22 <i>Track: Intelligent Factory Automation</i>
	 EDTConf 23 <i>General Track</i>
	 IFAC IMS 25 <i>Invited Session Track</i>
	 Universal Automation OPC UA Working Group
WG Member	

Research Publications

Conference Proceedings

- 1 **M. Majumder**, B. Wiesmayr, and A. Zoitl, "Digital Twin Architecture Patterns for Real-Time Control Systems," in *8th IEEE International Conference on Industrial Cyber-Physical Systems (ICPS)*, IEEE, 2025. [DOI: 10.1109/ICPS65515.2025.11087873](https://doi.org/10.1109/ICPS65515.2025.11087873).
- 2 **M. Majumder** and A. Zoitl, "A Conceptual Modelling Language for CPPS," in *30th IEEE International Conference on Emerging Technologies and Factory Automation (ETFA)*, IEEE, 2025. [DOI: 10.1109/ETFA65518.2025.11205675](https://doi.org/10.1109/ETFA65518.2025.11205675).
- 3 **M. Majumder** and A. Zoitl, "Model-as-a-Middleware: Information Model as an Integrator for Cyber-Physical Production Systems," in *23rd International Conference on Industrial Informatics (INDIN)*, In Press, IEEE, 2025.
- 4 **M. Majumder** and A. Zoitl, "Behaviour-oriented opc ua information model development for cpps," in *Model-Driven Engineering and Software Development*, F. J. Domínguez Mayo, L. F. Pires, and E. Seidewitz, Eds., Cham: Springer Nature Switzerland, 2024, pp. 54–74, ISBN: 978-3-031-66339-0.
- 5 **M. Majumder**, "A Domain-Driven Model Generation Framework for Cyber-Physical Production Systems," in *ACM/IEEE International Conference on Model Driven Engineering Languages and Systems Companion (MODELS-C)*, 2023, pp. 172–178. [DOI: 10.1109/MODELS-C59198.2023.00044](https://doi.org/10.1109/MODELS-C59198.2023.00044).
- 6 **M. Majumder**, B. Wiesmayr, and A. Zoitl, "Extending the OPC UA Companion Specification for an IEC 61499-based Control System," in *27th International Conference on Emerging Technologies and Factory Automation (ETFA)*, IEEE, 2023, pp. 1–4. [DOI: 10.1109/ETFA54631.2023.10275677](https://doi.org/10.1109/ETFA54631.2023.10275677).
- 7 **M. Majumder** and A. Zoitl, "A Domain-Driven Design Oriented OPC UA Server Development Methodology for CPPS," in *27th International Conference on Emerging Technologies and Factory Automation (ETFA)*, IEEE, 2023, pp. 1–8. [DOI: 10.1109/ETFA54631.2023.10275496](https://doi.org/10.1109/ETFA54631.2023.10275496).
- 8 **M. Majumder** and A. Zoitl, "A Modelling Methodology for Developing an Information Model for Cyber-Physical Production Systems using OPC UA," in *11th International Conference on Model-Based Software and Systems Engineering (MODELSWARD)*, INSTICC, SciTePress, 2023, pp. 152–159. [DOI: 10.5220/0011684900003402](https://doi.org/10.5220/0011684900003402).
- 9 **M. Majumder** and A. Zoitl, "A Proposal for OPC UA Companion Specification for IEC 61499 Based Control Application," in *19th International Conference on Factory Communication Systems (WFCS)*, IEEE, 2023, pp. 1–8. [DOI: 10.1109/WFCS57264.2023.10144247](https://doi.org/10.1109/WFCS57264.2023.10144247).
- 10 **M. Majumder** and A. Zoitl, "Run-time Configuration of the IEC 61499-based PLC-Service Bus via OPC UA," in *27th International Conference on Emerging Technologies and Factory Automation (ETFA)*, IEEE, 2023, pp. 1–4. [DOI: 10.1109/ETFA54631.2023.10275697](https://doi.org/10.1109/ETFA54631.2023.10275697).
- 11 V. Ashiwal, **M. Majumder**, and A. Zoitl, "Evaluation of Middleware Technologies for the PLC-Service bus in IEC 61499," in *IEEE 27th International Conference on Emerging Technologies and Factory Automation (ETFA)*, IEEE, 2022, pp. 1–4. [DOI: 10.1109/ETFA52439.2022.9921536](https://doi.org/10.1109/ETFA52439.2022.9921536).
- 12 **M. Majumder**, M. Shakil, and A. Zoitl, "Evaluation of PLC4X based Middleware as Integrator of Brownfield systems into Industrial Cyber-Physical Systems," 14th IFAC Workshop on

Intelligent Manufacturing Systems IMS 2022, 2022, pp. 427–432. [DOI: https://doi.org/10.1016/j.ifacol.2022.04.231](https://doi.org/10.1016/j.ifacol.2022.04.231).

- 13 S. S. P. Olaya, A. Winkel, M. Ehrlich, **M. Majumder**, A. Schupp, and M. Wollschlaeger, “CANopen Flying Master Over TSN,” in *Kommunikation und Bildverarbeitung in der Automation*, J. Jasperneite and V. Lohweg, Eds., Berlin, Heidelberg: Springer Berlin Heidelberg, 2022, pp. 245–256, ISBN: 978-3-662-64283-2.
- 14 **M. Majumder**, S. S. P. Olaya, M. Ehrlich, L. Wisniewski, J. Jasperneite, and M. Wollschlaeger, “Work-in-Progress: Semantic Knowledge Base as a Solution for Heterogeneous Industrial Network Management,” in *16th IEEE International Conference on Factory Communication Systems (WFCS)*, 2020, pp. 1–4. [DOI: 10.1109/WFCS47810.2020.9114450](https://doi.org/10.1109/WFCS47810.2020.9114450).
- 15 S. K. Panda, **M. Majumder**, L. Wisniewski, and J. Jasperneite, “Real-time Industrial Communication by using OPC UA Field Level Communication,” in *25th IEEE International Conference on Emerging Technologies and Factory Automation (ETFA)*, 2020, pp. 1143–1146. [DOI: 10.1109/ETFA46521.2020.9211998](https://doi.org/10.1109/ETFA46521.2020.9211998).
- 16 S. K. Panda, L. Wisniewski, M. Ehrlich, **M. Majumder**, and J. Jasperneite, “Plug & Play Retrofitting Approach for Data Integration to the Cloud,” in *16th IEEE International Conference on Factory Communication Systems (WFCS)*, 2020, pp. 1–8. [DOI: 10.1109/WFCS47810.2020.9114523](https://doi.org/10.1109/WFCS47810.2020.9114523).
- 17 F. Ansah, **M. Majumder**, H. de Meer, and J. Jasperneite, “Network Slicing : An Industry Perspective,” in *24th IEEE International Conference on Emerging Technologies and Factory Automation (ETFA)*, 2019, pp. 1367–1370. [DOI: 10.1109/ETFA.2019.8869073](https://doi.org/10.1109/ETFA.2019.8869073).
- 18 **M. Majumder**, L. Wisniewski, and C. Diedrich, “A Comparison of OPC UA & Semantic Web Languages for the purpose of Industrial Automation Applications,” in *24th IEEE International Conference on Emerging Technologies and Factory Automation (ETFA)*, 2019, pp. 1297–1300. [DOI: 10.1109/ETFA.2019.8869113](https://doi.org/10.1109/ETFA.2019.8869113).
- 19 S. K. Panda, **M. Majumder**, M. Ehrlich, A. Neumann, L. Wisniewski, and J. Jasperneite, “Topology Detection as a Base for Efficient Management of Heterogeneous Industrial Network Systems Using Software-Defined Networking,” in *15th IEEE International Workshop on Factory Communication Systems (WFCS)*, 2019, pp. 1–8. [DOI: 10.1109/WFCS.2019.8757933](https://doi.org/10.1109/WFCS.2019.8757933).